Astronomy and astrophysics – from the starry sky to the Big Bang

Astronomy is the oldest natural science and astrophysics one of the youngest. For thousands of years people measured the positions of the heavenly bodies; only in the 19th century did they begin to understand what they are made of and why they shine. Today we know that the stars are fusion reactors, that the elements in our bodies were made in them, that the universe is expanding from a hot beginning 13.8 billion years ago – and that 95 % of it consists of something we do not know. This page covers the field with three interactive figures and points to the site's many pages on the solar system, comets and eclipses.

Celestial mechanics – from Ptolemy to Newton

For centuries the Babylonians kept lists of the positions of the planets, and Greek astronomers built geometrical models. Ptolemy's *Almagest* (c. AD 150), with the Earth at the centre and the planets on epicycles, predicted the positions well enough to be used for 1400 years. In 1543 Copernicus put the Sun at the centre. On Hven Tycho Brahe measured the positions of the planets with the naked eye ten times more precisely than anyone before him, and from his Mars data in 1609–19 Kepler found his three laws: the orbits are ellipses, the radius vector sweeps out equal areas in equal times, and $T^2 \propto a^3$ (see Kepler's equation and the seasons).

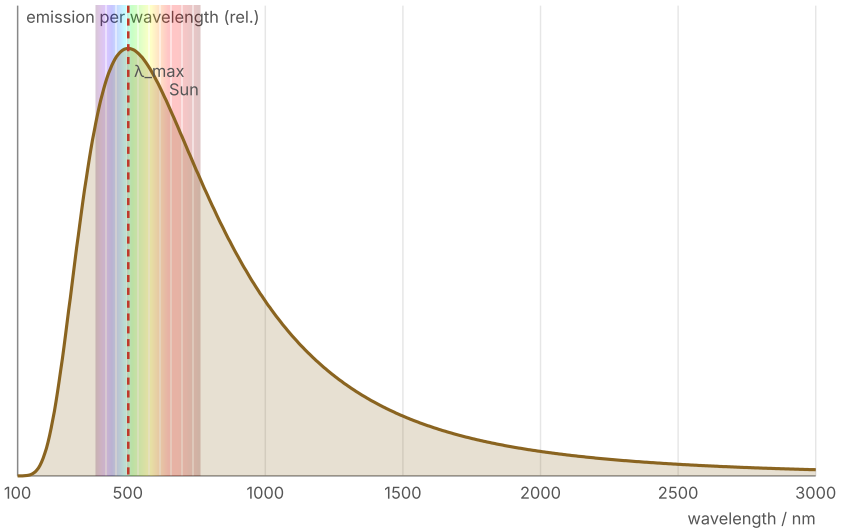
In 1609–10 Galileo turned a telescope on the sky and saw Jupiter's moons, the phases of Venus and the countless stars of the Milky Way. In the *Principia* (1687) Newton showed that Kepler's laws follow from a single universal law of gravitation (see Principia). Celestial mechanics became the science of triumph: Halley predicted the return of his comet, Gauss recovered the dwarf planet Ceres, and Le Verrier and Adams predicted Neptune, which was found in 1846 exactly where the calculation said. In 1676 the Dane Ole Rømer measured the finite speed of light from delays in the eclipses of Jupiter's moons.

Light from the sky

Almost everything we know about the universe comes from light – and since the 1930s from the whole electromagnetic spectrum: radio waves (Karl Jansky, 1932), microwaves, infrared, ultraviolet, X-rays and gamma rays, which can only be seen from space. A telescope's light-gathering power grows with its area, and its resolution with its diameter (see Resolving power). The James Webb Space Telescope (2021) has a 6.5 m mirror in the infrared; the Extremely Large Telescope in Chile will have 39 m.

A star radiates roughly like a black body. Planck's radiation law gives the spectrum, Wien's displacement law its peak and the Stefan–Boltzmann law the total emission:

$$\lambda_{\max} = \frac{2.898\ \text{mm K}}{T}, \qquad L = 4\pi R^2\,\sigma T^4$$



The spectral lines reveal the composition (see Atomic physics). Around 1900 the Harvard astronomers – Williamina Fleming, Antonia Maury and above all Annie Jump Cannon – classified hundreds of thousands of stellar spectra into the classes O, B, A, F, G, K, M. In her PhD thesis of 1925 Cecilia Payne showed that the classes are a temperature sequence, and that the stars consist mainly of hydrogen and helium – a result so surprising that she was persuaded to call it "almost certainly not real". The Doppler shift of the lines gives the velocity of the stars along the line of sight.

Distances – the cosmic distance ladder

Parallax

As the Earth orbits the Sun, nearby stars shift slightly relative to distant ones. In 1838 Friedrich Bessel measured the first parallax, of 61 Cygni: 0.3″. One parsec is the distance at which the parallax is 1″ (3.26 light years). The Gaia satellite has measured parallaxes for almost two billion stars.

Cepheids

In 1912 Henrietta Leavitt discovered that pulsating Cepheids in the Small Magellanic Cloud are brighter the longer their period. The Dane Ejnar Hertzsprung calibrated the law in 1913. Cepheids could then be used as “standard candles” in other galaxies.

Supernovae

Type Ia supernovae – exploding white dwarfs – have almost the same peak brightness and can be seen across billions of light years.

Redshift

For distant galaxies the redshift z is itself a measure of distance via Hubble’s law $v = H_0 d$. See The Hubble tension.

The physics of stars

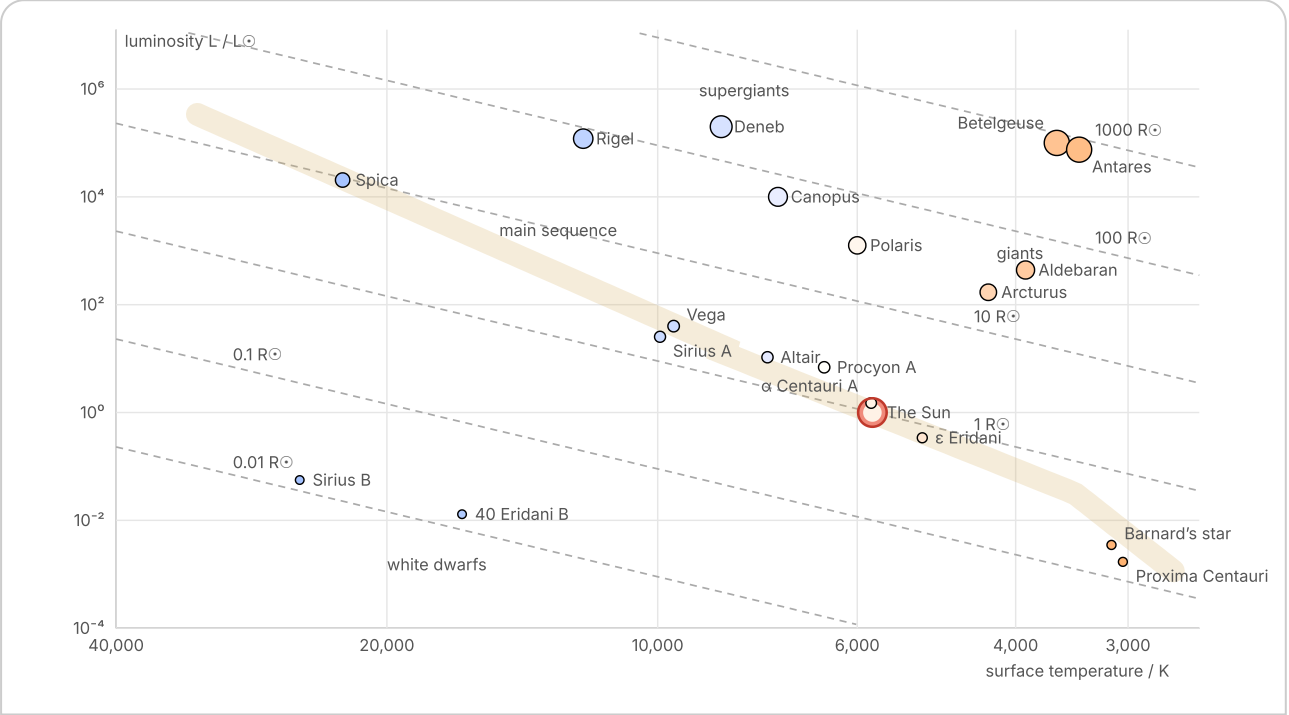
Why does the Sun shine? In the 19th century Kelvin and Helmholtz proposed that it contracts and converts gravitational energy into heat – which would give about 30 million years, far too little for the geologists and Darwin. In 1920 Arthur Eddington proposed that the Sun converts hydrogen into helium, and that the mass defect of 0.7 % provides the energy. In 1938–39 Hans Bethe and Carl Friedrich von Weizsäcker worked out the reactions: the pp chain in the Sun and the CNO cycle in heavier stars (see Nuclear physics). Neutrinos from the Sun’s core were measured by Ray Davis from 1968 – and the fact that there were too few was only resolved in 2001 with the discovery of neutrino oscillations.

A star is a ball of gas in balance between gravity and pressure – *hydrostatic equilibrium*. Energy is carried outwards by radiation and convection, so slowly that a photon from the Sun’s core takes tens of thousands of years to reach the surface. In *The Internal Constitution of the Stars* (1926) Eddington laid the foundations of stellar models, and in the 1930s Bengt Strömgren in Copenhagen developed them further. A star’s mass determines almost everything: luminosity, temperature, lifetime and fate.

$$\frac{dP}{dr} = -\frac{G M(r)\,\rho(r)}{r^2}, \qquad L \propto M^{3.5}, \qquad t_{\mathrm{life}} \approx 10^{10}\ \mathrm{yr}\cdot\left(\frac{M}{M_\odot}\right)^{-2.5}$$

The Hertzsprung–Russell diagram

In 1905–07 the Danish chemical engineer and astronomer Ejnar Hertzsprung noticed that stars of the same spectral class can have very different luminosities: there are giants and dwarfs. In 1913 Henry Norris Russell plotted luminosity against spectral class for many stars. The diagram became the most important tool of astrophysics: most stars lie on the *main sequence*, where they burn hydrogen in their cores, while giants, supergiants and white dwarfs are stars in other phases of life.



The life and death of stars

When the hydrogen in the core is used up, the core contracts and heats up while the outer layers expand: the star becomes a red giant. At 100 million degrees helium fuses to carbon and oxygen. Stars like the Sun end by puffing off their outer layers as a planetary nebula, and the core becomes a *white dwarf* the size of the Earth, held up by the Fermi pressure of the electrons. In 1930 Chandrasekhar showed that a white dwarf can weigh at most 1.4 solar masses.

Massive stars continue fusing up to iron. Then there is no more energy to be had (see the binding-energy curve in Nuclear physics), the core collapses in a fraction of a second, and the star explodes as a *supernova*. What remains is a *neutron star* – an atomic nucleus 20 km across – or a *black hole*. In 1967 Jocelyn Bell Burnell found the first pulsars, rotating neutron stars. Supernova 1987A in the Large Magellanic Cloud was detected in neutrinos a few hours before the light. Tycho Brahe's nova of 1572 – a supernova – showed that the starry sky is not unchanging.

The elements heavier than hydrogen and helium were made in stars and scattered by supernovae and neutron-star mergers. We are, as Carl Sagan put it, made of star stuff.

The Milky Way and the galaxies

In 1918 Harlow Shapley showed from the distribution of globular clusters that the Sun is not at the centre of the Milky Way but about 26,000 light years from it. In the “Great Debate” of 1920 Shapley and Heber Curtis argued over whether the spiral nebulae were gas clouds in the Milky Way or distant “island universes”. Edwin Hubble settled it in 1924 by finding Cepheids in the Andromeda nebula: it lay far outside the Milky Way. The universe suddenly consisted of billions of galaxies. At the centre of the Milky Way lies a black hole of four million solar masses, Sagittarius A*, whose existence Reinhard Genzel and Andrea Ghez proved by following the orbits of stars (Nobel Prize 2020). See Galaxy formation and Galaxy collisions.

Cosmology – the history of the universe

In 1927 Georges Lemaître showed that Einstein’s equations give an expanding universe and predicted that the redshift of galaxies grows with distance. Hubble found the relationship observationally in 1929. In 1931 Lemaître proposed that the universe began as a “primeval atom” – the idea that became the Big Bang. In 1948 George Gamow and his students predicted a relic of the hot beginning: radiation at a few kelvin. Arno Penzias and Robert Wilson found it in 1965 as noise in a radio antenna – the *cosmic microwave background* at 2.725 K. COBE (1992) measured its perfect Planck spectrum and the tiny temperature fluctuations from which the galaxies have grown; WMAP and Planck mapped them with great precision.

Big Bang theory also predicts that the first minutes of the universe produced about 25 % helium and traces of deuterium and lithium – exactly as observed. The universe is 13.8 billion years old. The theory of inflation (Alan Guth, 1980) proposes that in the very first fraction of a second the universe expanded exponentially; it explains why the universe is so uniform and flat, but has not yet been confirmed directly.

Dark matter and dark energy

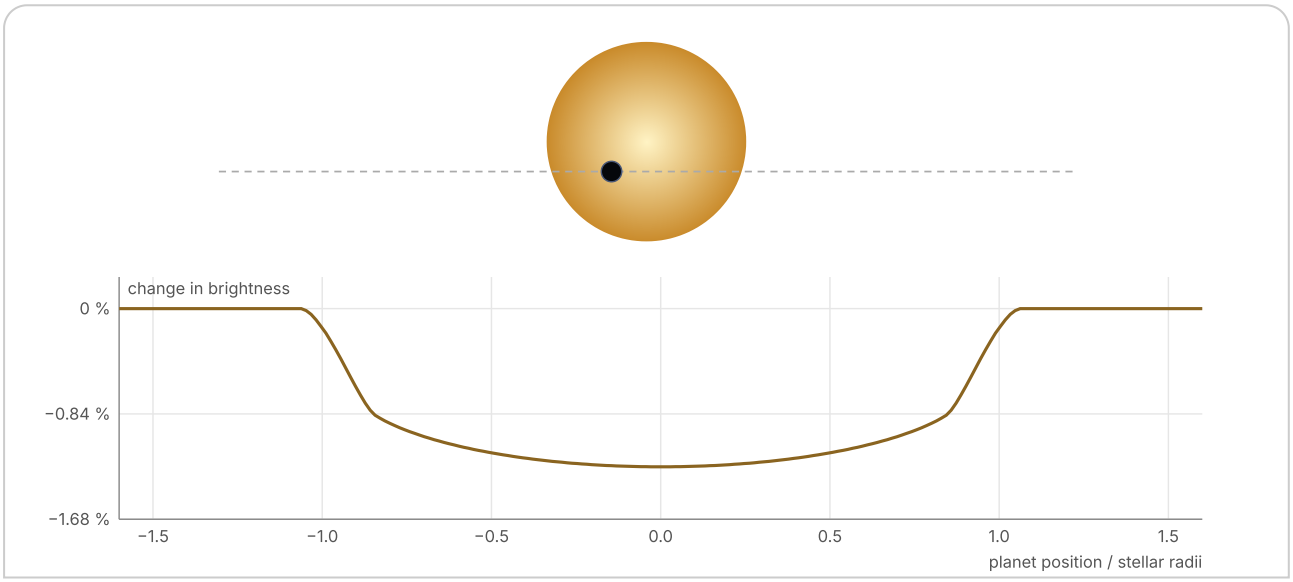
In 1933 Fritz Zwicky found that the galaxies in the Coma cluster move so fast that the cluster must contain far more mass than can be seen. In the 1970s Vera Rubin and Kent Ford showed that stars and gas in the outer parts of galaxies orbit just as fast as those further in – the rotation curves are flat. If Kepler and Newton are to hold, the galaxies must be surrounded by a halo of invisible *dark matter*. Gravitational lensing and the microwave background confirm that there is about five times as much dark matter as ordinary matter. What it is, nobody knows (see Is physics in crisis?).

In 1998 two teams, led by Saul Perlmutter and by Brian Schmidt and Adam Riess, used distant supernovae to find that the expansion of the universe is *accelerating* (Nobel Prize 2011). The cause is called *dark energy* and makes up about 68 % of the energy of the universe. The

simplest explanation is Einstein's cosmological constant, but the DESI measurements hint that it may change with time.

Exoplanets

In 1995 Michel Mayor and Didier Queloz found the first planet around a Sun-like star, 51 Pegasi b, by measuring the star's wobble with the Doppler method (Nobel Prize 2019). It was a "hot Jupiter" in a four-day orbit – something nobody had expected. The Kepler satellite (2009–18) and TESS found thousands of planets with the *transit method*: when a planet passes in front of its star, the brightness drops slightly. Today more than 6000 exoplanets are known, and the James Webb Space Telescope measures their atmospheres.



New messengers

Today astronomers observe more than light. Neutrinos from the Sun and from supernova 1987A, cosmic rays with energies far above the LHC's and – since 2015 – gravitational waves from merging black holes and neutron stars (see Relativity). On 17 August 2017 the gravitational wave GW170817 was followed by a gamma-ray burst and a kilonova seen by 70 observatories – the birth of *multi-messenger* astronomy. The Event Horizon Telescope has shown the shadows of black holes. And amateur astronomers still contribute: with comets, variable stars and photographs (see Astrophotos and Comet orbits).

Further reading

- Bradley W. Carroll and Dale A. Ostlie: *An Introduction to Modern Astrophysics* (2nd ed., Cambridge 2017)
- Michael A. Hoskin (ed.): *The Cambridge Concise History of Astronomy* (1999)
- Dava Sobel: *The Glass Universe* (2016) – on the women of the Harvard Observatory

- Steven Weinberg: *The First Three Minutes* (1977)
- Helge Kragh: *Cosmology and Controversy* (Princeton 1996)
- The biographies of Tycho Brahe, Kepler, Hubble, Lemaître, Chandrasekhar and Hawking

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